

Lactic Fermented Dairy Products (LFDP™)

Module: Lactic Fermented Dairy Products (LFDP™)
Standard: Natural Fermented Beverage Standard
Category: Product & System
Subcategory: Fermented Biological Systems
Type: Process Integrity Module

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Compatibility: Origin Eco™ · Product Integrity Standards ·
Natural Systems

Canonical Definition

Lactic Fermented Dairy Products define a class of milk-based products transformed through lactic acid fermentation.

A product is valid only if its structure, function, and identity are primarily created through biological fermentation rather than artificial modification.

A. Module Abstract

LFDP™ establishes the minimum conditions under which dairy-based products may be classified as true fermentation-derived systems.

It distinguishes products defined by biological transformation from those defined by chemical processing or additive-driven structure.

B. Core Principle

Fermentation must define the product.

C. Core Conditions

1. Biological Transformation

The product must result from active lactic fermentation.

2. Process Dominance

Fermentation must be the primary mechanism shaping structure, function, and identity.

3. Non-Simulation

The product must not rely on artificial structuring or chemical

processes that replace fermentation.

4. Additive Integrity

Additives may be present only if they do not override, simulate, or replace fermentation as the defining mechanism.

5. Safety

The product must be microbiologically stable and safe for consumption.

D. Prohibited Conditions

The product is invalid if:

- fermentation is absent or secondary
 - chemical or mechanical processes define the product instead of fermentation
 - additives simulate fermentation outcomes
 - the system is structurally artificial
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E. Scope

Applies to all dairy systems derived from lactic fermentation, including both liquid and solid forms.

This includes yogurt systems, kefir systems, and all milk-based fermentation processes.

F. Scope Extension Rule

Although the parent standard is defined for fermented beverages, the biological process defined in this module applies equally to liquid and solid forms derived from the same fermentation system.

G. MIF — Minimum Implementation Framework

A system implementing LFDP™ must:

- use milk-based input material
 - enable natural lactic fermentation
 - ensure fermentation is the dominant transformation process
 - maintain controlled fermentation conditions
 - verify that fermentation defines the final product
 - prevent artificial simulation of fermentation
 - ensure additives do not redefine product identity
 - ensure safety and stability of the final system
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H. System Output

A valid LFDP™ system produces a product that is:

- biologically transformed
- fermentation-defined in structure and function
- stable and safe
- not artificially simulated

I. Use Cases

Use Case 1 — Clean Yogurt Production

A producer uses natural fermentation to create yogurt. Structure and taste are derived from fermentation. The product remains valid under LFDP™.

Use Case 2 — Additive-Dominated Product

A product labeled as yogurt relies on stabilizers, sweeteners, and processing to define its structure. Fermentation is secondary. The product is reclassified and does not meet LFDP™ conditions.

Use Case 3 — Kefir-Based System

A kefir product is produced using natural fermentation processes. The microbial system defines the product. The product remains valid under LFDP™.

Use Case 4 — Long-Term Market Drift

Over time, products labeled as fermented become dominated by additives. Fermentation becomes symbolic rather than functional. LFDP™ exposes this shift and invalidates classification as a true fermented system.

Canonical Closing Statement

Fermentation is not an ingredient. It is the process that defines the product. If fermentation does not define the product, the product is not a fermented system.

Closing Insight

The most valuable food systems are not engineered. They are preserved, understood, and allowed to function as living processes.

Related Documents

→ [Natural Fermented Beverage Standard \(NFBS™\)](#)

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